

Chapter 16

Quantum Is a Public Good

Translating quantum innovation into solutions for business and society

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16.1 Professional Foundation

Looking back on my journey into finance, what stands out is that it was never driven by a primary interest in markets or investments. It was rather, from the beginning, an intellectual journey shaped by a fascination for mathematics and for the possibility of understanding complex systems through formal reasoning. In other words, I came to finance somewhat by detour. What attracted me was not the financial industry itself, but the opportunity it offered to apply mathematical tools to real-world problems involving uncertainty, risk, and decision-making.

After graduating from one of the leading French business schools, I felt a strong desire to pursue graduate studies in mathematics and statistics and eventually enrolled in a PhD in finance at UC Berkeley, where I was exposed to a rigorous vision of financial economics. I had the great privilege of having the late Mark Rubinstein as my PhD advisor. Mark was not only an exceptionally deep financial economist – perhaps best known for the Cox–Ross–Rubinstein discrete-time option pricing model, among many other outstanding contributions – but also a man of remarkable culture and intellectual breadth. To this day, I remember him once telling me that his favorite book in his entire (and extensive) library was a French book, *The Count of Monte Cristo* by the French author Alexandre Dumas. He ranked it even above Shakespeare’s plays, for which he had the greatest admiration.

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The early years of my academic career in the United States, at the University of Southern California at Los Angeles, followed by my return to EDHEC, were decisive in shaping my perspective. Very early on, I became convinced that finance, as a discipline, suffered from a structural tension: on the one hand, a strong theoretical framework grounded in elegant mathematical models; on the other, a real-world environment characterized by frictions, behavioral biases, and institutional constraints. Over the years, I became increasingly aware of the limitations of an ivory-tower approach to finance—one that emphasizes academic publication over engagement with practitioners, whose perspectives are critical to identifying and addressing real-world investment challenges.

This conviction led me to co-found the EDHEC-Risk Institute, with a very clear ambition: to contribute to reshaping best investment practices through the development of what I have often called “useful finance”—finance that is not only theoretically sound, but also practically relevant. Over time, this vision translated into a number of concrete initiatives that had a tangible impact on the asset management industry. One of the most significant contributions was our work on factor investing and smart beta strategies. At a time when capitalization-weighted indices were still largely dominant, we developed a research agenda aimed at providing efficient and robust alternatives based on well-documented risk premia.

This work ultimately led to the creation of Scientific Beta, a venture designed to bridge academic research and industrial implementation. Through this academic work and related industrial initiative, we were able to influence industry practices in a meaningful way—promoting a more disciplined, evidence-based approach to investment, and contributing to the broader adoption of smart factor investing as a more efficient way to harvest risk premia in equity markets.

At the core of my work has always been a focus on investment solutions and a paradigm that could be described as “precision finance”—a framework in which portfolios are no longer built through the mass distribution of standardized products, but through the design of tailored investment solutions aligned with specific investor objectives, constraints, and risk profiles. This shift implies moving away from a product-centric logic, often driven by marketing and scale considerations, toward a solution-oriented approach grounded in economic relevance and robustness. In such a framework, the focus is not on selling pre-packaged exposures, but on engineering portfolios that efficiently deliver desired outcomes—whether in terms of income generation, risk control, or long-term capital preservation—while explicitly accounting for uncertainty, estimation risk, and implementation constraints. This includes issues such as retirement investing, risk management, and long-term asset allocation.

My recent appointment as Research Director at the CFA Institute Research Foundation also provides me with a unique vantage point on the evolution of the global investment industry. It has reinforced my conviction that finance is at a turning point, where traditional approaches are increasingly challenged by technological transformations.

At the same time, this reflection on the evolving nature of finance has also led me to revisit a longstanding intellectual thread in my own journey which has been running in parallel to my academic career in finance: a deep and persistent fascination for

physics. For a long time, this remained a personal pursuit—something I explored outside of my professional life. In retrospect, it is clear that this dual interest in finance and physics was not entirely accidental. Both fields are concerned, in their own ways, with understanding complex systems under uncertainty through suitable mathematical structures.

16.2 The Quantum Pivot

If there was a “pivot” toward quantum physics and quantum technologies, it did not happen overnight. It was the result of a long intellectual maturation, and perhaps also of a form of fidelity to an earlier, more fundamental curiosity about the nature of the physical world that I have developed as a child.

A key turning point occurred during my sabbatical at Princeton in 2011-2012. Immersed in an environment immensely rich in the history of theoretical physics, with Albert Einstein, John Von Neumann or Robert Oppenheimer among the leading figures, I found myself reconnecting with a long-standing intellectual attraction. Attending two doctoral-level courses one in general relativity, and the other in cosmology by Paul Steinhardt remains, to this day, the most intellectually exhilarating experience of my career.

This experience led me to pursue a PhD in astrophysics, and later to participate in the international LIGO collaboration that achieved the first detection of gravitational waves. Being part of this effort, even at a modest level, was an extraordinary privilege—an experience that profoundly reshaped my relationship to science.

But the real turning point came later, during a subsequent sabbatical at MIT in 2022-2023, where I had the privilege of interacting with, and learning, from exceptional scholars such as Seth Lloyd at MIT or Jacob Barandes at Harvard, and where I became deeply immersed in the foundations of quantum mechanics.

My work focused in particular on the problem of time in quantum systems, and more specifically on what is known as the time-of-arrival problem—a fundamental question that, surprisingly, remains unresolved within the standard formalism of quantum mechanics. This led to a series of publications in leading physics journals, including work on time-of-arrival distributions for both continuous and discrete quantum systems, and the development of a general framework to address these questions without departing from the standard quantum formalism.

What fascinated me in this line of research was not only the technical challenge, but the conceptual depth. The fact that such a basic question—“when does a particle arrive?”—does not have a universally accepted answer within quantum mechanics reveals something profound about the limits of our current understanding of the physical world. At the same time, these questions are not purely abstract. They have direct implications for quantum information processing, including quantum computation and quantum communication, where the notion of time plays a critical role in algorithmic execution and information transfer. Different tasks such as circuit execution times, error detection latencies, or communication and synchronization

protocols, may naturally call for different notions of arrival time.

While my work has been primarily foundational so far, there are several concrete domains in which operational notions of time-of-arrival distributions may play a substantive role, for instance in applications to information propagation in locally interacting quantum systems. The so-called Lieb-Robinson bounds¹ rigorously constrain the maximal velocity of information spreading but they provide worst-case envelopes rather than operational access times. Arrival time distributions associated with specific detection protocols could instead characterize when excitations or logical information become experimentally accessible at a remote site. In this sense, these concepts complement velocity bounds by introducing a probabilistic notion of information arrival time, which may help enhance available estimates based on circuit depth time performance in many body systems.

It was at this point that the connection between my work in physics and my background in finance became evident. On the one hand, I was working on foundational questions in quantum theory and applications in quantum technologies. On the other, I had spent decades studying complex optimization and decision problems in finance and their applications in financial engineering. The realization was that we are entering a new phase—the second quantum revolution—where these foundational insights are beginning to translate into technological capabilities. Quantum computing is not just about faster computation. It is about a fundamentally different way of representing and manipulating information. And this has immediate and profound implications for fields like finance. In that sense, the “spark” was not a single moment, but the gradual realization that two intellectual trajectories I had pursued independently for many years were, in fact, converging toward a common frontier.

From this journey that has allowed a somewhat dreamy teenager to reconcile a passion for physics with a professional career in finance, I take away one central lesson: the importance of remaining, in some sense, faithful to one’s childhood dreams. This fidelity strikes me as a valuable guide in the seemingly modest yet extraordinary adventure that is a human life. Quantum physics teaches us that, at the heart of the fabric of the universe, lies a vast wave function—a generative source of infinitely many possibilities. In much the same way, human life can be seen as a continuous journey of self-transcendence, growth, realization, and the unfolding of these latent potentials. Just as it takes time, vision, talent, and sustained effort for an artist to transform what exists only in potential—clay for a sculptor, a blank page for a writer, or a blank canvas for a painter—into reality, each of us must build our own destiny step by step, assuming full responsibility for being the sole architect of this long and patient process through which our life ultimately takes shape. All of this is beautifully captured in Nietzsche’s famous injunction: “Become who you are. Do what only you can do.”

¹Lieb, E.H.; Robinson, D.W. The finite group velocity of quantum spin systems. *Commun. Math. Phys.* 1972, 28, 251-257.

16.3 Identifying the Void

The creation of the EDHEC Quantum Institute emerged from a very clear observation: despite the rapid acceleration of quantum technologies, there is a profound gap between technological development and economic understanding.

On the one hand, we are witnessing massive investments, strong governmental support, and rapid progress in quantum hardware and algorithms. On the other hand, there is a striking lack of clarity among business leaders, policymakers, and even academics about what quantum mechanics tells us about the world, the physical world, and how quantum technologies are going to impact our economies.

“Shut up and calculate” is not a viable option. This famous expression, due to David Mermin, reflects a crude pragmatic injunction according to which physicists should simply apply the unambiguous mathematical rules of quantum mechanics to obtain correct predictions, instead of trying to interpret the meaning of the theory. While this approach has proven remarkably effective in advancing the field, it deliberately sets aside deeper questions about what quantum mechanics actually tells us about the nature of reality. And for the development of quantum technologies, such an attitude is no longer sufficient. If these technologies are to fulfill their promise, they must be understood—not mystified. They are not based on magic, but on our ability to harness the logic of the physical world at its most fundamental level. What may appear counterintuitive is not irrational; it reflects a deeper structure of reality that we are only beginning to grasp.

In that sense, we are at a stage comparable to that of our ancestors when they first learned to master fire. We are aware of its transformative potential, we can generate sparks, and we are beginning to experiment with its applications. But we are still learning how to control it, stabilize it, and deploy it safely and effectively at scale. And we need to be able to communicate and explain to non-physicists what these new technologies are all about. The challenge ahead is therefore not only technological, but also conceptual. It requires moving beyond fascination and toward understanding—building a coherent framework that allows scientists, engineers, and decision-makers alike to engage with quantum technologies in a rigorous and informed manner. Only then can we transition from isolated demonstrations to robust, real-world impact.

In other words, while physics and engineering departments are doing an amazing job at pushing the boundaries of our knowledge in quantum science and technologies, there is a lack of institutions capable of translating quantum innovation into business-relevant insights. There is also a need for independent analysis to distinguish between what is feasible today, what is plausible in the medium term, and what remains purely aspirational.

In finance, this gap is particularly problematic. The industry is both a potential beneficiary of quantum technologies and a potential victim, especially in the context of cybersecurity and post-quantum cryptography. It is somewhat difficult for decision makers in the financial industry to understand these threats and opportunities, and how best to prepare for these technological advances. There is a certain tendency,

which we call “quantum washing”, to attribute near-term economic or operational benefits to quantum technologies either through implicit assumptions that are not currently satisfied or by focusing on problems chosen primarily for their amenability to quantum advantage rather than for their genuine economic relevance. Claims of universal speedups or systematic outperformance that often mix long-term theoretical possibilities with present-day capabilities makes it particularly challenging for non-experts to assess the reality of what a genuine quantum advantage can mean.

Beyond these technical considerations, there is a deeper issue: the lack of quantum literacy among decision-makers. This is where I believe business schools have a critical role to play. Training engineers and physicists is, of course, essential. But it is equally important to train business leaders who are capable of understanding the strategic implications of quantum technologies—who can ask the right questions, make informed decisions, and avoid both excessive skepticism and blind enthusiasm.

This represents a fascinating and non-trivial challenge. How do you teach quantum concepts—often perceived as abstract, counterintuitive, even “mysterious”—to students with a business background? How do you strike the right balance between conceptual rigor and accessibility?

This challenge is at the heart of the EDHEC Quantum Institute, which aims at bridging quantum science and real-world impact to unlock transformative solutions for business and society by leveraging EDHEC existing areas of expertise and existing strategic partnerships. We have launched research programs on foundations of quantum mechanics, quantum technologies in finance and insurance, and quantum technologies in information and communication, in partnership with EURECOM, an internationally renowned academic center for research and education in digital sciences. In a second step, we envision expanding the scope to other business domains, such as quantum technologies for AI, for the new space industry or for the biotech industry, which are already strongly impacted by the 2nd quantum revolution.

The Institute was designed as a bridge structure, with four core pillars: research, education, innovation, and outreach. Its ambition is obviously not to compete with physics departments, but to operate at the interface between science and application. The education component is central. The talent gap in quantum technologies is large and new educational programs are needed to train experts that can combine knowledge in quantum physics and business applications. In a nutshell, the education pillar of the EDHEC Quantum Institute is designed to equip future business leaders, engineers, and policymakers with the foundational knowledge and strategic insight required to navigate the rapidly evolving landscape of quantum technologies. By integrating quantum science with management education, the initiative addresses the growing need for interdisciplinary talent capable of bridging deep tech and real-world applications. For this, we develop programs that demystify quantum technologies, provide a clear conceptual framework, and connect these concepts to real-world use cases.

16.4 The Future Landscape

Quantum computing is widely expected to bring major improvements to financial problem-solving. In the area of portfolio optimization, quantum algorithms such as the Quantum Approximate Optimization Algorithm (QAOA) can solve complex construction problems involving high-dimensional constraints more efficiently than classical algorithms. In risk analysis and financial simulations, quantum-enhanced Monte Carlo methods offer the potential for faster and more accurate results, especially in pricing derivatives and calculating Value at Risk. Additionally, quantum machine learning (QML) is being explored as a means to detect patterns in large financial datasets, enabling better fraud detection, credit risk assessment, and trading strategy development. Quantum communication also has significant implications for finance. Quantum Key Distribution (QKD) can enable ultra-secure communication, which is critical for transmitting sensitive financial data. As quantum computers evolve, they will also pose a threat to existing encryption systems. Preparing for this shift through post-quantum cryptography will be essential to ensure data privacy and security in future financial infrastructures.

In the near term, the use of quantum computing in finance will be focused on hybrid approaches, combining classical and quantum methods. The objective for the time being is not to achieve a clear quantum advantage, but to build capabilities, identify use cases, and develop an operational understanding of the technology. In the medium term, as hardware improves, we can expect more significant advances in areas such as optimization, simulation, and machine learning. At the same time, the emergence of quantum threats to existing cryptographic systems will force a transition toward post-quantum infrastructures.

EDHEC has a long-standing reputation for academic leadership and research in data-driven finance, which offers a solid foundation for engaging in quantum finance. The EDHEC Quantum Institute is developing research on portfolio construction, asset pricing, and risk management, areas where quantum technologies are expected to have the most impact.

Establishing a dedicated research cluster focused on quantum finance is also essential to establish thought leadership. As the quantum finance field is still in its early stages, we ambition to influence the direction of research and standard-setting within the industry. As a thought leader, we also ambition to play a key role in policy development and industry regulation. The ethical, legal, and social implications of quantum technologies are still being defined. Through informed research and engagement, the EDHEC Quantum Institute can contribute to establishing responsible frameworks for their use in finance.

But beyond these technical developments, I believe the most important challenge is conceptual. Quantum mechanics is often presented as mysterious, counterintuitive, even incomprehensible. This narrative, while appealing, is ultimately counterproductive. It creates barriers to understanding and opens the door to confusion—and sometimes even to forms of “quantum mysticism” that have no scientific basis.

As already mentioned, one of our key ambitions at the EDHEC Quantum Institute is

therefore to clarify the concepts, and help economic decision makers understand that quantum mechanics, while profoundly different from classical physics, is not magical. It is based on a coherent logical structure, and it can be understood—at least to a meaningful extent—by non-specialists. This effort of clarification is essential, not only for educational purposes, but also for the development of the field itself. A discipline cannot mature if its foundational concepts remain obscure to those who are expected to use it. In this context, we intend to explore the possibility of developing certification frameworks, which would allow professionals to demonstrate a certain level of understanding of quantum technologies and their business implications. Such certifications could play a role similar to what the CFA designation has played in finance—providing a common language, a shared set of standards, and a signal of competence.

The EDHEC Quantum Institute ambitions to develop onsite and online executive programs aimed at professionals seeking to understand the strategic implications of quantum disruption. In parallel, we are developing an Innovation Hub with a dedicated quantum track within the TechForward incubator created by EDHEC and EURECOM, which is meant to support early-stage ventures that apply quantum technologies to business, finance, communication, security, health, logistics, and other industries. By connecting startups to cutting-edge research from EDHEC and EURECOM, the incubator aims to foster ventures grounded in both scientific rigor and commercial relevance. A special track is devoted to student-led startups emerging from the Institute's master and MBA programs.

In a way, the EDHEC Quantum Institute is an attempt to address a fundamental inefficiency: the fact that technological revolutions often outpace our collective ability to understand and integrate them. More broadly, our ambition is to contribute to the structuring of the emerging field of quantum finance. This involves not only developing new methods and applications, but also defining standards, best practices, and evaluation frameworks. It also means engaging with industry, policymakers, and the broader public to ensure that the development of quantum technologies is aligned with societal needs.

Again, my core belief is that quantum is a public good, too important to be reserved for a small circle of initiated experts. Its implications reach far beyond physics, shaping industry, policy, security, and education. In addition to equipping us to engage with emerging quantum technologies, engagement with quantum principles offers a further intellectual dividend, one that is particularly useful at a time when public discourse is increasingly shaped by simplistic, one-sided views that distort and caricature what is inherently subtle and layered. Quantum theory is not only a powerful scientific and technological framework; it is also a welcome celebration of complexity, an invitation to contemplate reality and to understand that only a multiplicity of perspectives allows us to grasp its full richness.